

Auteur: Noran Ben Said
Studierichting: Informatica
Oefeningenreeks 5A nr. 7.4

$$\begin{aligned} & \int_0^{\pi} \cos^2 x \sin^3 x \, dx \\ &= \int_0^{\pi} \cos^2 x \sin^2 x \sin x \, dx \\ &= \int_0^{\pi} \cos^2 x (1 - \cos^2 x) \sin x \, dx \end{aligned}$$

Stel $t = \cos x \Rightarrow dt = -\sin x \, dx$

Nieuwe grenzen:

Als $x = 0$, dan $t = \cos 0 = 1$

Als $x = \pi$, dan $t = \cos \pi = -1$

$$\begin{aligned} &= \int_1^{-1} t^2 (1 - t^2) (-dt) \\ &= \int_{-1}^1 t^2 (1 - t^2) \, dt \\ &= \int_{-1}^1 (t^2 - t^4) \, dt \\ &= \int_{-1}^1 t^2 \, dt - \int_{-1}^1 t^4 \, dt \\ &= \left[\frac{t^3}{3} \right]_{-1}^1 - \left[\frac{t^5}{5} \right]_{-1}^1 \\ &= \left(\frac{1}{3} - \left(-\frac{1}{3} \right) \right) - \left(\frac{1}{5} - \left(-\frac{1}{5} \right) \right) \\ &= \frac{2}{3} - \frac{2}{5} = \frac{4}{15} \end{aligned}$$

References